

Consistent gyroscopic velocity in the tangent contact equations

The adapter now includes both bodies' implicit gyroscopic angular-velocity increments in each tangent RHS. Upstream normal equations and final body writeback already include them. The signed torque Jacobian for body B supplies its own minus sign; an extra sign must not be inserted.

A native two-body regression chooses each COM velocity so that the free contact point velocity, including gyro, is zero. In upstream tangents the two spurious RHS components are approximately 0.481 and -1.742 m/s; the corrected components are below 1e-12 m/s. The normal component is already zero. This regression exercises anisotropic inertia and gyroscopic increments on both bodies; it is included in `spatial_friction_checks` and hosted builds.

The correction fixes the mechanics mismatch. It **does not resolve every nonlinear contact solve**. A separately frozen follow-up replays the two rejected random-hull scenes with identical authored geometry, mass, material coefficients, gravity, prescribed wall motion, timestep ladder and acceptance gates. Every one of the six attempts still rejects the strict 1e-8 m/s contact residual gate:

Case	Travel fraction	Final rejected residual (m/s)
Eight shaking hulls, seed 42	0.06	1.77895e-8
Eight shaking hulls, seed 42	0.03	1.08727e-8
Eight shaking hulls, seed 42	0.015	3.47786e-5
27 rotating/shaking hulls, seed 7301	0.06	2.30410e-5
27 rotating/shaking hulls, seed 7301	0.03	1.71763e-7
27 rotating/shaking hulls, seed 7301	0.015	1.97331e-5

There is no trajectory history or verified fast setting for a rejected run. Reasons, timings and complete inputs are retained; no friction-pyramid fallback is accepted. Unchanged gates make this a disclosed remaining solver/integration limit, not a claimed successful random-shape engine. Further active-set handling of redundant contacts and a verified compliant contact branch remain necessary.

Execution source: 78e63313b5d13dd80170d03c54bcaffccbbf47e8. Baseline source: 7c279768f6518e66b1ee620d44bd9b37309. The exact source snapshot, six rejection receipts and scenes are SHA-256-pinned in `results/summary.json`. `python -m research.audit_spatial_gyro` independently checks source integrity, equality of authored physics and controls, every retained attempt and the absence of any qualification. The original 76-attempt study and its three qualified references remain unchanged.